

```

program finalCodeExample
{
  declare A,B;

  procedure proc(in a, inout b)
  {
    declare c;
    function func()
    {
      declare d;
      d:=4;
      A:=B;
      b:=a;
      return(c+d)
    }

    # body of proc #
    c:=3;
    print(A);
    print(B);
    B:=func();
    print(A);
    print(B)
  }

  # body of finalCodeExample #
  A:=1;
  B:=2;
  call proc(in A, inout B);
  print(A);
  print(B)
}.

```

```

-----> entity A: 12 added to symbol table
-----> entity B: 16 added to symbol table
-----> entity a: in 12 added to symbol table
-----> entity b: inout 16 added to symbol table
-----> entity procedure proc added to symbol table
-----> entity c: 20 added to symbol table
-----> entity function func added to symbol table
-----> entity d: 12 added to symbol table
1: begin_block, func, _, _
2: :=, 4, _, d
3: :=, B, _, A
4: :=, a, _, b
-----> entity T_1: 16 added to symbol table
5: +, c, d, T_1
6: return, T_1, _, _
7: end_block, func, _, _

```

Symbol table at the end of func

=====

A: 12 -- B: 16 -- procedure proc: None, None, ['a: in 12', 'b: inout 16']

a: in 12 -- b: inout 16 -- c: 20 -- function func: 1, 20, []

d: 12 -- T_1: 16

=====

```
.data
str_nl: .asciz "\n"
.text
```

L0:

```
# 0: jump, _, _, main
    j Lmain
```

L1:

```
# 1: begin_block, func, _, _
    sw ra, (sp)
```

L2:

```
# 2: :=, 4, _, d
    li t1, 4
    sw t1, -12(sp)
```

L3:

```
# 3: :=, B, _, A
    lw t1, -16(gp)
    sw t1, -12(gp)
```

L4:

```
# 4: :=, a, _, b
    lw t0, -4(sp)
    addi t0, t0, -12
    lw t1, (t0)
    lw t0, -4(sp)
    addi t0, t0, -16
    lw t0, (t0)
    sw t1, (t0)
```

L5:

```
# 5: +, c, d, T_1
    lw t0, -4(sp)
    addi t0, t0, -20
    lw t1, (t0)
    lw t2, -12(sp)
    add t1, t2, t1
    sw t1, -16(sp)
```

L6:

```
# 6: return, T_1, _, _
    lw t1, -16(sp)
    lw t0, -8(sp)
    sw t1, (t0)
```

L7:

```
# 7: end_block, func, _, _
```

```

        lw ra, (sp)
        jr ra
    8: begin_block, proc, _, _
    9: :=, 3, _, c
    10: print, A, _, _
    11: print, B, _, _
-----> entity T_2: 24 added to symbol table
    12: par, T_2, ret, func
    13: call, func, _, _
    14: :=, T_2, _, B
    15: print, A, _, _
    16: print, B, _, _
    17: end_block, proc, _, _

```

Symbol table at the end of proc

=====

A: 12 -- B: 16 -- procedure proc: 8, 28, ['a: in 12', 'b: inout 16']
a: in 12 -- b: inout 16 -- c: 20 -- function func: 1, 20, [] -- T_2:
24

=====

L8:

```

# 8: begin_block, proc, _, _
    sw ra, (sp)

```

L9:

```

# 9: :=, 3, _, c
    li t1, 3
    sw t1, -20(sp)

```

L10:

```

# 10: print, A, _, _
    lw a0, -12(gp)
    li a7, 1
    ecall
    la a0, str_nl
    li a7, 4
    ecall

```

L11:

```

# 11: print, B, _, _
    lw a0, -16(gp)
    li a7, 1
    ecall
    la a0, str_nl
    li a7, 4
    ecall

```

L12:

```

# 12: par, T_2, ret, func
    addi fp, sp, 20
    addi t0, sp, -24
    sw t0, -8(fp)

```

```

L13:
# 13: call, func, _, _
      sw sp,-4(fp)
      addi sp,sp,20
      jal L1
      addi sp,sp,-20

L14:
# 14: :=, T_2, _, B
      lw t1,-24(sp)
      sw t1,-16(gp)

L15:
# 15: print, A, _, _
      lw a0,-12(gp)
      li a7,1
      ecall
      la a0,str_nl
      li a7,4
      ecall

L16:
# 16: print, B, _, _
      lw a0,-16(gp)
      li a7,1
      ecall
      la a0,str_nl
      li a7,4
      ecall

L17:
# 17: end_block, proc, _, _
      lw ra,(sp)
      jr ra
18: begin_block, main, _, _
19: :=, 1, _, A
20: :=, 2, _, B
21: par, A, in, proc
22: par, B, inout, proc
23: call, proc, _, _
24: print, A, _, _
25: print, B, _, _
26: halt, _, _, _
27: end_block, main, _, _

```

Symbol Table at the end of the main program:

=====

A: 12 -- B: 16 -- procedure proc: 8, 28, ['a: in 12', 'b: inout 16']

=====

```

L18:
Lmain:
# 18: begin_block, main, _, _
      addi sp,sp,20

```

```

        mv gp, sp

L19:
# 19: :=, 1, _, A
        li t1, 1
        sw t1, -12(gp)

L20:
# 20: :=, 2, _, B
        li t1, 2
        sw t1, -16(gp)

L21:
# 21: par, A, in, proc
        addi fp, sp, 28
        lw t0, -12(gp)
        sw t0, -12(fp)

L22:
# 22: par, B, inout, proc
        addi t0, sp, -16
        sw t0, -16(fp)

L23:
# 23: call, proc, _, _
        sw sp, -4(fp)
        addi sp, sp, 28
        jal L8
        addi sp, sp, -28

L24:
# 24: print, A, _, _
        lw a0, -12(gp)
        li a7, 1
        ecall
        la a0, str_nl
        li a7, 4
        ecall

L25:
# 25: print, B, _, _
        lw a0, -16(gp)
        li a7, 1
        ecall
        la a0, str_nl
        li a7, 4
        ecall

L26:
# 26: halt, _, _
        li a0, 0
        li a7, 93
        ecall

L27:
# 27: end_block, main, _, _

```